MSO uniformisation over trees

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1 Trees

Definition 1.1 (Tree)**.** A *tree* over an alphabet Σ is recursively defined as follows:

- a, where $a \in \Sigma$.
- $a(t, t')$, where $a \in \Sigma$ and t, t' are trees.

The set of all trees over Σ is Tree_Σ

Definition 1.2 (Nodes)**.** The *nodes* of a tree are defined as follows:

$$\mathrm{Nodes}\,(t) = \begin{cases} \varepsilon & \text{if } t = a \\ \varepsilon \cup \{0u \mid u \in \mathrm{Nodes}\,(t')\} \cup \{1u \mid u \in \mathrm{Nodes}\,(t'')\} & \text{if } t = a(t', t'') \end{cases}$$

Definition 1.3 (Leaves)**.** The *leaves* of a tree are defined as follows:

$$\mathrm{Leaves}\,(t) = \begin{cases} \varepsilon & \text{if } t = a \\ \{0u \mid u \in \mathrm{Leaves}\,(t')\} \cup \{1u \mid u \in \mathrm{Leaves}\,(t'')\} & \text{if } t = a(t', t'') \end{cases}$$

Remark 1.0.1. $\mathit{Leaves}\,(t) \subseteq \mathit{Nodes}\,(t)$

Definition 1.4. We note $t\,(n)$ the *label* of the node n in the tree t :

$$t\,(n) = \begin{cases} a & \text{if } t = a \text{ and } n = \varepsilon \\ a & \text{if } t = a(t', t'') \text{ and } n = \varepsilon \\ t'\,(m) & \text{if } t = a(t', t'') \text{ and } n = 0m \\ t''\,(m) & \text{if } t = a(t', t'') \text{ and } n = 1m \end{cases}$$

Definition 1.5. Let t be a tree and $x, y \in \mathrm{Nodes}\,(t)$, we define the *ancestor* relation as

$$x \sqsubseteq y \text{ if and only if } \exists z \in \mathrm{Nodes}\,(t)\,, y = xz$$

2 Tree Automata

Definition 2.1 (*Non-Deterministic Bottom-Up Tree automaton*)**.** A Non-Deterministic Bottom-Up Tree automaton (NBUA) is defined as a tuple $(\Sigma, Q, I, \Delta, F)$ where:

- Σ is an alphabet.
- Q is a finite set of *states*.
- $I \subseteq \Sigma \times Q$ correspond to the possible states of the leaves.
- $\Delta \subseteq Q \times Q \times \Sigma \times Q$ is the *transition relation*.
- $F \subseteq Q$ is the set of *final states*.

Definition 2.2 (Run of an automaton)**.** A *run* ρ of a NBUA A over a tree t is :

$$\begin{array}{rcl} \rho : \text{Nodes}\,(t) & \to & Q \\ (t\,(b)\,, \rho(n)) \in I & \text{if} & n \in \text{Leaves}\,(t) \\ (\rho(n0), \rho(n1), t\,(n)\,, \rho(n)) \in \Delta & \text{if} & n \in \text{Nodes}\,(t) \setminus \text{Leaves}\,(t) \end{array}$$

We say that ρ is *acceptant* if $\rho(\varepsilon) \in F$ and A accepts t if $F \cap \{\rho(\varepsilon) \mid \rho \text{ run of } A \text{ over } t\} \neq \emptyset$

Definition 2.3 (Language of an automaton)**.** Let A be a NBUA. Its *associated language* is defined as:

$$\mathcal{L}\,(A) = \{t \in \text{Tree}_\Sigma \mid \text{exists } \rho \text{ a accepting run of } A \text{ over } t\}$$

Definition 2.4 (*Deterministic Bottom-Up Tree automaton*)**.** A Deterministic Bottom-Up Tree automaton (DBUA) is NBUA that that verifies

$$\forall q, r, a, t, t' \;\; (q, r, a, t) \in \Delta \wedge (q, r, a, t') \in \Delta \implies t = t'$$

Remark 2.0.1. *A DBUA can also be defined as as a tuple $(\Sigma, Q, init, \delta, F)$ where:*

- *Σ is an alphabet.*
- *Q is a finite set of states.*
- *$init : \Sigma \to Q$ is a function that initializes the states of the leaves.*
- *$\delta : Q \times Q \times \Sigma \to Q$ is the transition map.*
- *$F \subseteq Q$ is the set of final states.*

We will show that the two definitions are equivalent.

Definition 2.5 (Interpretation of an automaton)**.** The *interpretation* of a DBUA A over an alphabet Σ is defined as follows:

$$\begin{array}{rcl} [\![A]\!] : \text{Tree}_\Sigma & \to & Q \\ a & \mapsto & \text{init}_a \\ a(t, t') & \mapsto & \delta_a([\![A]\!]\,(t)\,, [\![A]\!]\,(t')) \end{array}$$

Remark 2.0.2. *In the case of a DBUA there exists a unique run that corresponds to :*

$$\begin{array}{rcl} \rho : \mathit{Nodes}\,(t) & \to & Q \\ n & \mapsto & \mathit{init}_{t(n)} \ \text{if } n \in \mathit{Leaves}\,(t) \\ n & \mapsto & \delta_{t(n)}(\rho(n0), \rho(n1)) \ \text{if } n \in \mathit{Nodes}\,(t) \setminus \mathit{Leaves}\,(t) \end{array}$$

Lemma 2.6. *Let A be a DBUA,*

$$\mathcal{L}\,(A) = \{t \in \mathit{Tree}_\Sigma \mid [\![A]\!]\,(t) \in F\}$$

Proof. We will prove by induction on a tree t that (a) there exists a run of A over t, and (b) for all runs ρ of A over t, $[\![A]\!]\,(t) = \rho(\varepsilon)$. We proceed by case distinction.

- If $t = a$. Let ρ be defined by $\rho(\varepsilon) = \mathrm{init}_a$, then ρ is a run of A over t. We have proved (a). Consider now some run ρ of A over t. We have $\rho(\varepsilon) = \mathrm{init}_a = [\![A]\!]\,(a) = [\![A]\!]\,(t)$. We have proved (b).

- If $t = a(t', t'')$. By induction hypothesis (a), there exists runs ρ' and ρ'' on t' and t'' respectively. Let ρ be defined by $\rho(\varepsilon) = \delta_{t(\varepsilon)}(\rho'(\varepsilon), \rho''(\varepsilon))$, $\rho(0u) = \rho'(u)$ for all $u \in \mathrm{Nodes}\,(t')$ and $\rho(1u) = \rho''(u)$ for all $u \in \mathrm{Nodes}\,(t'')$. Then ρ is a run of A over t. We have proved (a).

  Consider now some run ρ of A over t. Let $\rho'(u) = \rho(0u)$ for all $u \in \mathrm{Nodes}\,(t')$ and $\rho''(u) = \rho(0u)$ for all $u \in \mathrm{Nodes}\,(t'')$. Then ρ' is a run of A over t' and ρ'' over t''. By induction hypothesis (b) twice, we know that $\rho'(\varepsilon) = [\![A]\!]\,(t')$ and $\rho''(\varepsilon) = [\![A]\!]\,(t'')$. We have now $\rho(\varepsilon) = \delta_{t(\varepsilon)}(\rho'(\varepsilon), \rho''(\varepsilon)) = \delta_{t(\varepsilon)}([\![A]\!]\,(t'), [\![A]\!]\,(t'')) = [\![A]\!]\,(t)$. We have proved (b).

□

Definition 2.7. Let $A = (\Sigma, Q, I, \Delta, F)$ be a NBUA, we define the *power set automaton* of A, noted $\mathrm{Det}\,(A)$ as $\mathrm{Det}\,(A) = (\Sigma, \mathcal{P}\,(Q), \mathrm{init}, \delta, F')$ where:

- $\mathrm{init} : \Sigma \to \mathcal{P}\,(Q)$
  $a \mapsto \{q \in Q \mid (a, q) \in I\}$

- $\delta : \mathcal{P}\,(Q) \times \mathcal{P}\,(Q) \times \Sigma \to \mathcal{P}\,(Q)$
  $(X, Y, a) \mapsto \{q \in Q \mid (x, y, a, q) \in \Delta, x \in X, y \in Y\}$

- $F' = \{X \in \mathcal{P}\,(Q) \mid X \cap F \neq \emptyset\} \subseteq \mathcal{P}\,(Q)$

Theorem 2.8. *For all $N = (\Sigma, Q, I, \Delta, F)$ an NBUA, the power set automaton verifies that*

$$[\![\mathit{Det}\,(N)\,]\!]\,(t) = \{\rho(\varepsilon) \mid \rho \ \textit{run of } N \ \textit{over } t\} \ \textit{ and } \mathcal{L}\,(N) = \mathcal{L}\,(\mathit{Det}\,(N))$$

Proof. We will first show that for all tree t,

$$[\![\mathrm{Det}\,(N)\,]\!]\,(t) = \{\rho(\varepsilon) \mid \rho \ \mathrm{run} \ \mathrm{of} \ N \ \mathrm{over} \ t\} \ .$$

We will prove this property by induction over t. We proceed by case distinction.

- If $t = a$, then

$$\begin{aligned} \{\rho(\varepsilon) \mid \rho \text{ run of } N \text{ over } t\} &= \{q \in Q \mid (a, q) \in I\} \\ &= [\![\text{Det}\,(N)\,]\!]\,(t) \ . \end{aligned}$$

- If $t = a(t_1, t_2)$, then by the induction hypothesis we know that $[\![\text{Det}\,(N)\,]\!]\,(t_i) = q_i$ if and only if there exists a run over t_i ending at q_i, for $i \in \{1, 2\}$.

$$\begin{aligned} & q \in [\![\text{Det}\,(N)\,]\!]\,(t) \\ \text{if and only if} \quad & q \in \delta_a([\![\text{Det}\,(N)\,]\!]\,(t_1)\,, [\![\text{Det}\,(N)\,]\!]\,(t_2)) \\ \text{if and only if} \quad & q \in \{q \in Q \mid (x_1, x_2, a, q) \in \Delta, x_1 \in [\![\text{Det}\,(N)\,]\!]\,(t_1)\,, x_2 \in [\![\text{Det}\,(N)\,]\!]\,(t_2)\} \\ \text{if and only if} \quad & \exists x_1, x_2, (x_1, x_2, a, q) \in \Delta, x_1 \in [\![\text{Det}\,(N)\,]\!]\,(t_1)\,, x_2 \in [\![\text{Det}\,(N)\,]\!]\,(t_2) \\ \text{if and only if} \quad & \text{there exists a run over } t_i \text{ ending at } x_i \text{ for } i \in \{1, 2\} \text{ and } (x_1, x_2, a, q) \in \Delta \\ \text{if and only if} \quad & q \in \{\rho(\varepsilon) \mid \rho \text{ run of } N \text{ over } t\} \end{aligned}$$

Then $[\![\text{Det}\,(N)\,]\!]\,(t) = \{\rho(\varepsilon) \mid \rho \text{ run of } N \text{ over } t\}$.

We must now show that $\text{Det}\,(N)$ accepts t if and only if N accepts t.

$$\begin{aligned} N \text{ accepts } t \quad & \text{if and only if} \quad F \cap \{q \in Q \mid (a, q) \in I\} \neq \emptyset \\ & \text{if and only if} \quad F \cap [\![\text{Det}\,(N)\,]\!]\,(t) \neq \emptyset \\ & \text{if and only if} \quad [\![\text{Det}\,(N)\,]\!]\,(t) \in F' \\ & \text{if and only if} \quad \text{Det}\,(N) \text{ accepts } t \end{aligned}$$

and so $\mathcal{L}\,(N) = \mathcal{L}\,(\text{Det}\,(N))$.

TODO: Discuss proof □

Definition 2.9. An NBUA of transition relation Δ is *order-insensitive* if for all transitions $(p, q, a, r) \in \Delta$, $(q, p, a, r) \in \Delta$.

Remark 2.0.3. *For a DBUA of transition map δ, it is order-insensitive if and only if $\delta_a(p, q) = \delta_a(q, p)$ for all states p, q and letter a.*

Lemma 2.10. *If $N = (\Sigma, Q, I, \Delta, F)$ a NBUA is order-insensitive, then Det(N) is also order-insensitive.*

Proof.

$$\begin{aligned} \delta_a(X, Y) &= \{q \in Q \mid (x, y, a, q) \in \Delta, x \in X, y \in Y\} \\ &= \{q \in Q \mid (y, x, a, q) \in \Delta, x \in X, y \in Y\} \quad (N \textit{ is order-insensitive}) \\ &= \delta_a(Y, X) \end{aligned}$$

□

2.1 MSO

Definition 2.11. Let $t \in \text{Tree}_\Sigma$, we define a model of t as :

$$\mathcal{M}(t) := (\text{Nodes}(t), \sqsubseteq, (a(x))_{a\in\Sigma})$$

where : $a(x)$ if and only if $t(x) = a$

We will discuss some examples of MSO formulas.

Example 2.1.1. The formula $\text{leaf}(x) = \forall y, y \sqsubseteq x$ encodes the fact that x is a leaf.

Example 2.1.2. x is the root of the tree is equivalent to :

$$\text{root}(x) = \forall y, x \sqsubseteq y$$

Example 2.1.3. The following formula encodes the fact that x and y are the immediate children of z:

$$\text{children}(x, y, z) = x \neq y \wedge z \sqsubseteq x \wedge z \sqsubseteq y \wedge \forall w\, ((z \sqsubseteq w \wedge (w \sqsubseteq x \vee w \sqsubseteq y)) \to (w = x \vee w = y))$$

Lemma 2.12. *If A is a order-insensitive NBUA then there exists a MSO formula ϕ such that for all $t \in \text{Tree}_\Sigma$,*

$$\mathcal{M}(t) \models \phi \text{ if and only if } t \in \mathcal{L}(A)$$

Proof. Let $A = (\Sigma, Q, I, \Delta, F)$ be a NBUA, and $k = |Q|$.

$$\phi = \exists X_0 \ldots \exists X_{k-1} \qquad \forall x \left(\bigvee_{i=0}^{k-1} X_i(x) \right) \tag{1}$$

$$\wedge \quad \forall x \left(\bigwedge_{0\leq i\leq j\leq k-1} \neg(X_i(x) \wedge X_j(x)) \right) \tag{2}$$

$$\wedge \quad \forall x \left(\text{leaf}(x) \to \bigvee_{(a,i)\in I} X_i(x) \right) \tag{3}$$

$$\wedge \quad \forall x \forall y \forall z \left(\text{children}(x, y, z) \to \bigvee_{(p,q,a,r)\in\Delta} (X_p(x) \wedge X_q(y) \wedge a(z) \wedge X_r(z)) \right) \tag{4}$$

$$\wedge \quad \forall x \left(\text{root}(x) \to \bigvee_{q\in F} X_q(x) \right) \tag{5}$$

TODO: fix overfull
$j \in X_i$ if and only if , for a given run, the state i is reached from the node j.
The formula ϕ encodes the following properties:

1. Every node has an associated state.
2. A node cannot be in two states at the same time.

3. The transitions originate from the leaves.
4. The transitions of inner nodes ensure that for every trio of nodes, where two are children of the third, there exists a transition from the children to the parent.
5. Finally, the word is accepted if and only if the state associated with the root node is a final state.

From this we deduce that

$$\mathcal{M}(t) \models \phi \text{ if and only if } t \in \mathcal{L}(A)$$

□

Definition 2.13. The *cartesian product automata* of two NBUA $A_1 = (\Sigma, Q_1, I_1, \Delta_1, F_1)$ and $A_1 = (\Sigma, Q_2, I_2, \Delta_2, F_2)$ with final states $F' \subseteq Q_1 \times Q_2$ is :

$$A_1 \times A_2 = (\Sigma, Q_1 \times Q_2, I_1 \times I_2, \Delta_3, F')$$

where $\Delta_3 = \{((q_1, q_2), (p_1, p_2), (a_1, a_2), (r_1, r_2) \mid (q_1, p_1, a_1, r_1) \in \Delta_1, (q_2, p_2, a_2, r_2) \in \Delta_2\}$

Lemma 2.14. *Let $A_1 = (\Sigma, Q_1, I_1, \Delta_1, F_1)$ and $A_1 = (\Sigma, Q_2, I_2, \Delta_2, F_2)$ be two NBUA, then $A_3 = A_1 \times A_2$ with final states $F_1 \times F_2$ verifies that $\mathcal{L}(A_3) = \mathcal{L}(A_1) \cap \mathcal{L}(A_2)$*

Lemma 2.15. *Let $A_1 = (\Sigma, Q_1, I_1, \Delta_1, F_1)$ and $A_1 = (\Sigma, Q_2, I_2, \Delta_2, F_2)$ be two NBUA, then*

$$A_3 = (\Sigma, Q_1 \uplus Q_2, I_1 \uplus I_2, \Delta_1 \uplus \Delta_2, F_1 \uplus F_2)$$

verifies that $\mathcal{L}(A_3) = \mathcal{L}(A_1) \cup \mathcal{L}(A_2)$

Lemma 2.16. *Let $A = (\Sigma, Q, I, \Delta, F)$ the NBUA $A' = (\Sigma, Q, I, \Delta, Q \cap F)$ verifies that $\mathcal{L}(A') = (\mathcal{L}(A))^{\complement}$*

Lemma 2.17. *Over the alphabet $\Sigma \times \{0, 1\}$ the automaton A described bellow can decide the formula $a(x)$, where $a \in \Sigma$ and x is a free variable.*

$$A = (\Sigma \times \{0, 1\}, \{\mathit{Yes}, \mathit{No}\}, \mathit{init}, \delta, \mathit{Yes})$$

The following function definitions should be interpreted as a pattern match, with priority for the first rule :

- $\mathit{init} : \Sigma \times \{0, 1\} \to \mathcal{P}(Q)$
  $(a, 1) \mapsto \mathit{Yes}$
  $_ \mapsto \mathit{No}$
- $\delta : Q \times Q \times (\Sigma \times \{0, 1\}) \to Q$
  $(_, _, (a, 1)) \mapsto \mathit{Yes}$
  $(\mathit{No}, \mathit{No}, _) \mapsto \mathit{No}$
  $_ \mapsto \mathit{Yes}$

Lemma 2.18. *Over the alphabet $\Sigma \times \{0,1\}^2$ the automaton A described bellow can decide the formula $x \sqsubseteq y$, where x and y are free variables.*

$$A = (\Sigma \times \{0,1\}^2, \{\mathit{Yes},\ \mathit{No},\ \mathit{FoundY}\}, \mathit{init}, \delta, \mathit{Yes})$$

The following function definitions should be interpreted as a pattern match, with priority for the first rule :

- $\mathit{init} : \Sigma \times \{0,1\}^2 \to \mathcal{P}(Q)$
$$\begin{aligned} (_, _, 1) &\mapsto \mathit{FoundY} \\ _ &\mapsto \mathit{No} \end{aligned}$$
- $\delta : Q \times Q \times (\Sigma \times \{0,1\}^2) \to Q$
$$\begin{aligned} (\mathit{FoundY}, _, (_, 1, 0)) &\mapsto \mathit{Yes} \\ (_, \mathit{FoundY}, (_, 1, 0)) &\mapsto \mathit{Yes} \\ (\mathit{FoundY}, _, (_, 0, 0)) &\mapsto \mathit{FoundY} \\ (_, \mathit{FoundY}, (_, 0, 0)) &\mapsto \mathit{FoundY} \\ (\mathit{Yes}, _, _) &\mapsto \mathit{Yes} \\ (_, \mathit{Yes}, _) &\mapsto \mathit{Yes} \\ (_, _, (_, 0, 1)) &\mapsto \mathit{FoundY} \\ _ &\mapsto \mathit{No} \end{aligned}$$

Lemma 2.19. *If ϕ is a MSO formula there exists A a NBUA such that for all $t \in \mathit{Tree}_\Sigma$,*

$$\mathcal{M}(t) \models \phi \text{ if and only if } t \in \mathcal{L}(A)$$

Proof. The proof procedes by induction on the structure of ϕ.

If ϕ is an atomic formula, then the lemmas above decide it.

If ϕ is of the form $\psi_1 \wedge \psi_2$ then we can compute the intersection between the automata for ψ_1 and ψ_2, obtained using the induction hypothesis. We can deal with the disjuction using the union instead of the intersection.

If ϕ is of the form $\neg\psi$ then we can compute the complement of the automaton corresponding to ψ.

We have to deal with the case $\phi = \exists x.\psi(x)$. We can find a NBUA $A = (\Sigma, Q, I, \Delta, F)$ for ψ where x is treated as a free variable.

Then the automaton $A' = (\Sigma \times \{0,1\}, Q \times \{0,1\}, I \times \{0\}, \Delta', F \times \{1\})$ where

$$\Delta' = \left\{ \begin{array}{l} ((q,u),(q',u'),(a,X),(q'',u'')) \mid u, u' \in \{0,1\}, \\ \quad u'' = u \wedge u', X \text{ has } 0 \text{ at position } i, \\ \quad (q, q', (a, X), q'') \in \Delta \end{array} \right\}$$
$$\cup \left\{ \begin{array}{l} ((q,0),(q',0),(a,X),(q'',1)) \mid \\ \quad X3 \text{ has } 1 \text{ at position } i, \\ \quad (q, q', (a, X), q'') \in \Delta \end{array} \right\}.$$

and i corresponds to the variable x, can decide ϕ. If existential quantification is applied to a second-order variable, the automation remains unchanged. However, instead of considering a single variable in the transition function, look at all the variables within the given set.

$\square$